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Review article

A systematic review of open data in agriculture

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ABSTRACT

In this work, we perform a systematic literature review of Open Data and Public Domain datasets in Agriculture. We use the PRISMA method to analyze the existing academic literature about open data in agriculture, concretely 1401 papers from the IEEE Xplore and Web of Science collections, published from 2012 to 2022. Many of these articles use or make available datasets of very different typologies, like sensor data, statistical data or satellite images, among others kinds. Some papers talk about different relevant topics that influence the use of open data, like the lack of open data for research purposes, barriers to adopting or sharing data, privacy concerns, data-sharing recommendations and guidelines. In addition, with the help of a script created ad-hoc for this research work, we analyze the degree of compliance within the FAIR principles of all the public domain or open datasets that we have been able to identify from the literature, concretely 104 datasets. The script can check the compliance of Gen2 maturity indicators of a list of resources. Using these metrics, we have been able to identify those open datasets that might be at risk of stopping to be available and started a "rescue operation". For those datasets whose terms permitted it, we migrated those Open Data datasets to Zenodo, a repository that complies with the availability principles. This will ensure the future survival of valuable data in Agriculture.

1. Introduction

The current technological progress in the Computer Science area as well as other technological fields evidence the importance of data in our era. The exponential growth of data during the last years has opened up new opportunities and challenges for businesses and organizations across various industries, including Agriculture. The ability to collect, analyze, and interpret data has become a crucial factor in gaining a competitive edge, becoming resilient, and making informed decisions.

Every year, data is becoming a more valuable asset and there is a growing need for creating, handling, and managing data. Data is required in different fields related to digital agriculture such as, but not limited to, Artificial Intelligence, Data Analytics, Big Data, or Decision Support Systems. This situation requires a lot of effort from Institutions, Researchers, and companies, which have created different agreements, concepts, and standards to govern the data required to drive digital agriculture. According to Oliveira and Lóscio (2018-05-30), Béné (2022), nowadays, Data Ecosystems have emerged as a result of this need, where people and organizations share their data by using proper technology and resources.

In this context, Agriculture should not remain behind in this new trend of this sort of *Data fever*. Moreover, Agriculture is a strategic sector (Beckman and Countryman, 2021), and achieving food sustainability is a key concept nowadays, while some governments do not seem to have food sustainability as a priority (Kumar et al., 2022; Rose et al., 2022), for other institutions are making significant efforts towards sustainable systems (Davies, 2020-07-24; Lu et al., 2022; Rose et al., 2022). Concerning Data Ecosystems, nowadays some organizations do not share their data and there is a lack of standardized models and theories (Gelhaar et al., 2021).

In the Agriculture sector, there is a growing interest in generating and using data. We can find plenty of applications of data in Agriculture. For example, in the field of Big Data, we can mention the project CYBELE (Davy et al., 2020), funded by the European Union and coordinated by the South East Technological University of Ireland and participated by 31 research institutes from different EU countries. CYBELE aims to empower innovation and value generation in Agriculture by applying Big Data, Cloud Computing, and Internet of Things (IoT) technologies to improve Precision Agriculture and Precision Livestock Farming while offering services to different stakeholders like farmers, entrepreneurs or researchers. One of the objectives of this project is to mitigate the barriers to adopting digitalization. In this regard, the CYBELE project provides the aforementioned stakeholders with a platform for accessing datasets from various sources, as well as tools that enable them to experiment with the data. CYBELE is an excellent example of how to integrate cutting-edge technologies at different

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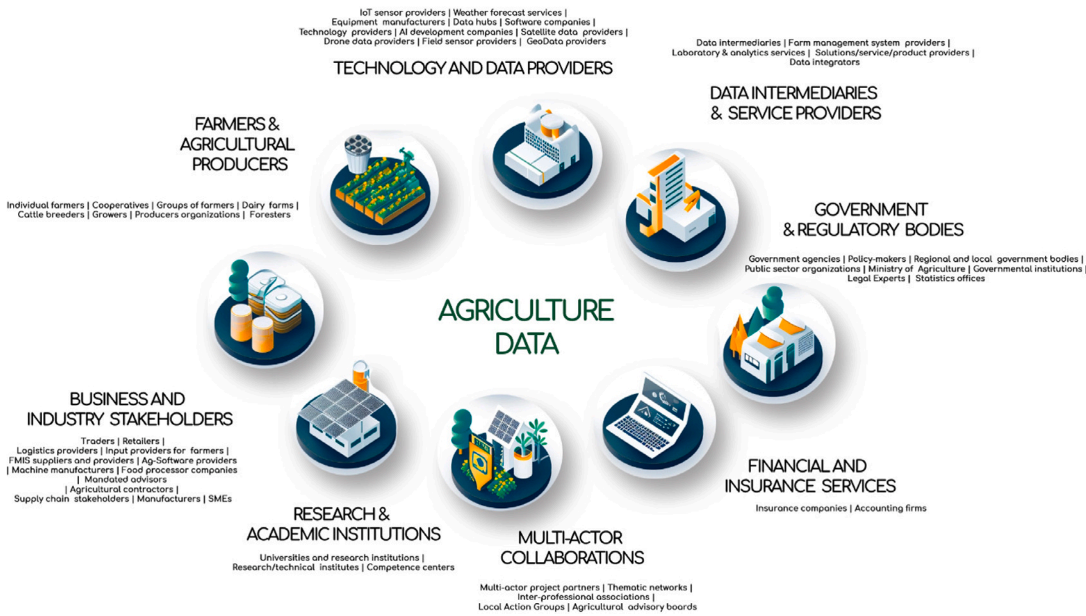

Fig. 1. Stakeholders in the Agriculture Data ecosystem.

levels (infrastructure, cloud services, Big Data). Perakis et al. (2020) introduces the project and analyzes twelve use cases. For example, protein content prediction on soya, predictive climate models and their impact on viticulture or pig weighing optimization. Montañana et al. (2020) also mentions other use cases like 3D farming implementation or developing algorithms to improve energy efficiency in agriculture.

GODAN (Global Open Data for Agriculture & Nutrition)[1] is an initiative and a voluntary association promoted by the United Nations and different governments to tackle the Zero Hunger Sustainable Development Goal number 2 by putting a special effort on open data in agriculture by embolden organizations and countries to develop open access policies and by empowering release and re-usability of data. Other relevant initiatives are CGIAR Platform for Big Data in Agriculture,[2] which applies Big Data approaches to agriculture to reduce poverty and to gain resilience against climate change, pest and diseases. The platform is becoming a broker of big data information and tries to share agriculture data. CGIAR is a non-profit research association funded by different governments and has different Research Centers across the world. The University of Minnesota auspices the GEMS Initiative,[3] which produces research data that applies the FAIR principles. GEMS has different case studies applied to the Agriculture field, we can mention some of them like improving irrigation decisions, detecting pet risks, or applying data to crop management. Regarding China, we can find different Open Data Initiatives, mainly at a regional level, as we can find open data platforms in different provinces like Zhejiang, Beijing, Qinghai or Sichuan, among others (Enbo and Na, 2020). In addition, the Chinese Government maintains the Data portal of the National Bureau of Statistics of China.[4] Nevertheless, the Chinese initiatives to empower the sharing and reusability of data are at an early stage, according to Wang (2022), Enbo and Na (2020) Andrade et al. (2021) describe different data-driven initiatives and highlights the need for ontologies and data sharing platforms. The authors describe the Crop Observatories, that try to harmonize the crop data to help in decision-making. Osinga et al. (2022) analyzes the current development status of each use case and proposes approaches to achieve them while applying Big Data techniques. Big Data can be applied to different topics, for example, Chaiklieng et al. (2022) proposes using big data to analyze farmers' accidents.

The use of data in agriculture affects different stakeholders: farmers and agricultural producers, business and industry stakeholders, research and academic institutions, multi-actor collaborations, financial and insurance services, government and regulatory bodies, data intermediaries and service providers or technology and data providers (see Fig. 1 for a detailed stakeholders map). Sometimes, there are conflicts of interest between these stakeholders. For example, Moritz et al. (2022) mentions that sometimes big industries can set barriers that establish difficulties for the rest of the stakeholders in their actions. Mehrabi et al. (2020), Nobes and Harris (2023) also identifies the lack of internet access for small farmers as an important barrier to adopting open data in Agriculture, this situation happens worldwide, but with a special impact on Africa.

Data is also a necessary asset for implementing the Internet of Things in agriculture. IoT helps to implement precision agriculture, which can help in the development of sustainable agri-food models. In that way, Ruan et al. (2019) propose an infrastructure and a model to facilitate data sharing. With the help of data and IoT, smart agriculture can be empowered, which results in an increment of productivity, as studied by Elijah et al. (2018), in which the authors identify the advantages of using data analytics in different agriculture application domains, like tracking, precision agriculture or greenhouse production.

Thanks to Machine Learning techniques, we can detect and model changes in groundwater level (Sahoo et al., 2017), extract relevant information from Agriculture databases (Amini and Rahmani, 2023) or from drone images (Fragassa et al., 2023).

But sharing data and building data ecosystems is not an easy task, and a lot of barriers and challenges should be overcome. For example, lack of transparency in ownership of data, sharing rights or privacy are barriers that make the farmers, especially the small ones, agnostic for sharing their data (Wiseman et al., 2019; Borrero and Mariscal, 2022). Some of these barriers can be overcome by making data available as Open Data, especially to favor these small players as described next. The authors of Shepherd et al. (2020) go into more detail by analyzing specific problems that pose barriers to adoption, from a lack of trust about the use and ownership of data (Who will be able to access the data generated by small farmers and for what purpose?), tensions in the economic model, with discrepancies of interests between public and

[1] https://www.godan.info (accessed on 25 February 2024).
[2] https://bigdata.cgiar.org/ (accessed on 25 February 2024).
[3] https://gems.umn.edu/ (accessed on 25 February 2024).
[4] https://data.stats.gov.cn/english (accessed on 25 February 2024).

private actors, and unclear benefits for farmers. To cite more examples, Steeneveld and Hogeveen (2015) identify the reasons why primary producers in the Netherlands are reluctant to invest in sensors. The most common reasons are that producers prefer to invest their money in other areas of their businesses and that they are unsure of the real usefulness of sensors. The authors also observe a correlation between farms with sensors, which tend to be larger and invest fewer hours of labor on average, and farms that do not adopt these technologies, which tend to be smaller and require more manual labor.

To address these barriers, there are several initiatives underway, many of which are discussed throughout this paper. We will now mention some initiatives that have been carried out in Africa, which are primarily focused on training, signing agreements and agreements to share open data, creating platforms, and investing in infrastructure (Mwelwa et al., 2020).

As it is not unreasonable to suppose, in Agriculture, there are strong differences in access, sharing, and generation of data between big farmers and small ones (Wiseman et al., 2019). In the academic literature, we can find efforts to mitigate these differences. For example, Borrero and Mariscal (2022) analyzes the use of data platforms on small farmers in Spain. The study reflects that small farmers want to participate in data-sharing platforms, especially if the platform is maintained by universities. Nevertheless, more effort should be paid in terms of transparency and privacy.

We can find another example focused on small fruit farmers in Colombia, where there is a digital gap between rural and urban areas. To reduce the gap, the Colombian government has different initiatives to reduce the digital gap between urban and rural zones as mentioned in Howland et al. (2015). The goal of this paper is to record the position of Colombian farmers regarding the data, to check their need for data, and the importance given to data sharing.

The significance of open access became evident in the 1990s when various initiatives emerged to encourage the scientific community to adopt measures facilitating the accessibility of science to a global audience. In this regard, we can cite the *ArXiv* repository,[5] which has been in existence since 1991. Additionally, in 1999, the Santa Fe Convention (United States) examined the potential of the web to enhance and streamline the communication of academic information (Costa and Leite, 2016). The Budapest Initiative for Open Access[6] is a declaration and a movement that promotes open access to scientific and academic information. It was launched in 2002 in Budapest and is promoted by a group of scientists and researchers. According to this initiative, the information and results should be freely available to any person who wishes to access them. The use of open licenses that allow reuse, redistribution, and adaptation of contents are important tools for achieving the Initiative goals, always recognizing the original authorship. The Budapest Initiative provides the following definition of Open Access: *By "open access" to this literature, we mean its free availability on the public internet, permitting any users to read, download, copy, distribute, print, search, or link to the full texts of these articles, crawl them for indexing, pass them as data to software, or use them for any other lawful purpose, without financial, legal, or technical barriers other than those inseparable from gaining access to the internet itself*.

The Budapest Initiative has held significant relevance, representing a notable advancement beyond the aforementioned milestones. This initiative has been extensively cited and referenced in documents, academic works, and various regulations. Furthermore, it aligns with subsequent declarations, such as the Berlin Declaration on Open Access to Knowledge in the Sciences and Humanities[7] (Costa and Leite, 2016). The Open Society Foundation is an international network that supports the implementation of the Budapest Initiative. In the year 2022, the Budapest Initiative released new recommendations to promote open access (Budapest Open Access Initiative, 2022). These new recommendations focus on Article Processing Charges (APCs), open-access journals, and research practices that incentivize open access. Therefore, the Budapest Initiative remains active to this day.

Janssen et al. (2012) performed a detailed summary of the benefits of adopting Open Data and the barriers that challenge the adoption of Open Data on different levels (Institutional, Task complexity, Legislation, Use and participation, Technical or Information Quality). It is not our goal to detail all these barriers, but we want to highlight some of them, like no uniform policies, lack of resources, data inaccuracy, privacy concerns, or accessibility problems, among others.

Crato (2023) discusses the current situation of Open Data culture, issues, and opportunities as well as strategies for solving open problems. For example, the authors mention strategies to preserve the anonymity of the data, which is an important barrier. Special mention should be made to the classification of datasets according to their origin (See Table 6.1 of Crato (2023), which is also based on Connelly et al. (2016)).

In this section, we have put over the table the importance of Data in the future of Agriculture and the Agrifood industry. In our work, we want to review the current status of Open Data Agriculture research. In particular, we want to check what data is being used, which papers generate open data, and what topics are the most frequent when using or producing Open Data. We want, in addition, to analyze the current status of Open Data and the level of maturity of data repositories that are publicly available. As an important amount of data is published under the Public Domain, we also decided to include these datasets. A creative work is under the Public Domain if it is no longer under intellectual property rights (Boyle, 2008). Consequently, these works are not owned by any individual and belong to the public.

Based on the situation regarding Open Data in agriculture identified so far, we want to focus our research on the following topics:

1. What are the public sources of Open Data used in Agriculture scientific literature?
2. Papers[8] which use data repositories.
3. Papers that generate Open Data.
4. Papers that use and generate data simultaneously.
5. What's the level of interoperability of datasets used in Agriculture research?
6. Fair Principles and Sustainable Development Goals (SDGs) related to Open Data.
7. Which of these datasets are at risk of disappearing, being no longer openly available?
8. What are the most relevant topics about Data management in Agriculture?

To achieve our goal, we have carried out a systematic review based on PRISMA (Page et al., 2021) of the academic literature we found on two relevant databases given the amount of research they publish in connection with the agriculture digitalization and data: Web of Science and IEEE Xplore. The rest of this paper is structured as follows:

1. Methods: describes the process of collecting, analyzing and reporting about the conducted systematic literature review using PRISMA.
2. Analysis: discusses the data collected and the patterns identified, like the most popular datasets, the types of licenses being used, compliance with FAIR Guiding principles, and other relevant topics found in the literature.
3. Conclusions: the final remarks about the study carried out in this paper.

5 https://arxiv.org/ (accessed on 25 February 2024).

6 https://www.budapestopenaccessinitiative.org/ (accessed on 25 February 2024).

7 https://openaccess.mpg.de/Berlin-Declaration (accessed on 25 February 2024).

8 Papers published in English during the last 10 years.

2. Methods

As previously introduced, our work is a review of the State of the Art of Open Data and Public Domain data sources in Agriculture research. Our systematic review follows the reporting guideline proposed by the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA 2020) (Page et al., 2021), which is designed to improve the quality of literature review processes as well as facilitate the replicability of the results. Down below, all the steps followed in the process are explained.

2.1. Information sources and search strategy

We obtained all the articles from two databases: Web of Science and IEEE Xplore. Both databases are well known by the scientific community and allow us to retrieve information from journals and conferences with high reputations. Web of Science was chosen because of its more general character: it incorporates information from journals of different editors and publishers. While IEEE Xplore contains papers more focused on technical fields from scientific and technological conferences, whose publications are also relevant to our research. In the beginning, we explored the possibility of including more databases, but the amount of information retrieved was very high to be handled and analyzed. The goal of this paper is not to create an exhaustive repository of papers and datasets related to Open Data in Agriculture but to provide an overall description regarding the State of The Art of the use of Open Data in this field. Including thousands of additional papers in our collection would probably generate a lot of unnecessary or redundant information. Finally, to set temporal windows, we collected the data during March and April 2023.

The first search used the term *agriculture* in combination with *datasets* and *open data*. This first approach was discarded because we found a lot of noise coming from the topic *Machine Learning* that was not related to data sharing. In addition, during this first review, we found that many articles about strategies or frameworks to facilitate data sharing and reproducibility mention the use of ontologies (Mughal et al., 2021b) for data integration. Therefore, we decided to include *ontology* as a query term while excluding *Machine Learning* from the search results.

Consequently, the search query for Web of Science is the following one:

TS = (agriculture data sharing) OR TS = (agriculture datasets) OR TS = (agriculture ontology) OR TS = (agriculture open data) NOT TS = (“Machine Learning”)

The query results are then filtered as follows to improve the relevance of the selected publications:

- Research area: Agriculture.
- Mesh heading: Agriculture.
- Language: English.
- Articles from 1 January 2012 to 31 December 2022.

In the case of IEEE Xplore, the query is adapted to the syntax expected by this database:

(“All Metadata”: agriculture data sharing) OR (“All Metadata”: agriculture datasets) OR (“All Metadata”: agriculture ontology) AND (“All Metadata”: agriculture open data) NOT (“All Metadata”: “Machine Learning”)

Then, as for Web of Science, the results are filtered to those from 2012 to 2022. In both cases, we chose that period to check the global trend and the evolution of agricultural data sharing. From 2012 because before that year, the literature was so scarce that going further does not contribute to the overall results of the analysis. And until 2022 because that way we can consider just complete years, and currently all literature published during 2022 is already in the considered indexes.

2.2. Eligibility criteria

2.2.1. Inclusion criteria

The criteria considered to include an article in the study are:

1. Being articles published in scientific journals, books, or conferences.
2. Regarding the content of their abstracts, either:
   (a) The abstract mentions the use or production of datasets, databases, or data collections.
   (b) It mentions strategies to generate, share, handle, or use data.
3. The article must include Agriculture as one of the most[9] relevant topics.

2.2.2. Exclusion criteria

1. The abstract mentions that the data is not directly available, it should be requested to the authors or it is not available.
2. The data used or generated is not free, there is some kind of cost associated.
3. The paper is not written in English.
4. The date of publication is before 2012 or after 2022.
5. The abstract does not explicitly mention the use of datasets, databases or data collections.

Note that we do not want to exclude non Open Data datasets in this initial stage, as we want to study differences and relations between them and Open Data datasets.

2.3. Selection process

After collecting the information of all papers retrieved from the two literature databases, a screening process was started to check if inclusion and exclusion criteria were met. We used the tool *Abstrackr* for performing the screening (Wallace et al., 2012).

The main goal of the process was to check if the title and the abstract of the article provide enough information to consider that the paper is using, sharing, or talking about data sources in Agriculture. As described in the exclusion criteria section, if there is no clear evidence of working with data sources, then, the paper is excluded.

2.4. Data collection

After selecting all the articles, we started the data collection process. During this stage, we identified all the data sources used in the papers by (a) Re-checking the abstract. (b) Checking the paper information page on the journal website. (c) Reading the full article. Note that, at this point, we only identify data sources, their analysis will be addressed later.

2.5. Data items

After reading and identifying the data sources, we initially defined three categories of papers:

1. Papers that use data sources related to the agriculture domain.
2. Papers whose authors created and shared this kind of data sources.
3. Papers that speak about existing agriculture-related data sources.

[9] To be considered as “most” relevant topic, the article should mention the word “Agriculture” or a derivative word at least in the title, the abstract or the keywords.

However, along the analysis process, it was necessary to add two additional categories for some special cases not covered by the initial ones:

1. Papers that discuss the lack of agriculture data sources and its implications.
2. Papers that mention traceability of data in agriculture research.

Each paper can be classified into more than one of the five categories that were finally used.

2.6. Risk of bias

To reduce the impact of bias during the process, articles are reviewed three times:

1. During the screening, by using *Abstrackr*.
2. During the Data collection stage, by reading the full article.
3. After being assigned to one of the five categories mentioned before. Each article is reviewed again to check that every identified dataset is available on the Internet.

2.7. Synthesis method

After identifying the data sources and classifying them into the previous categories, we performed two different analyses. First, a general analysis for all papers that comply with our inclusion criteria and contain general stats like countries, journals, topics, and other information about public data sources. Second, a more detailed analysis focusing on articles related to the most relevant data sources. We consider that a data source is relevant when it is referred from two or more articles.

2.8. Additional papers included in the review

Finally, to verify that most of the relevant publications have been already considered, we performed a search in Google Scholar with two different queries: *Open data in Agriculture*, and *Agriculture datasets*. Then, we cross-checked the results with those obtained so far from Web of Science and IEEE Xplore. The results were very satisfactory, showing the relevance of the selected sources of literature. We just identified six additional papers, that comply with the inclusion and exclusion criteria but were not already included. Consequently, we included them in the collection.

2.9. Results

A total of 1401 papers were collected at the first instance as a result of executing the query in the Web of Science and IEEE Xplore literature databases. After the first screening focusing just on their abstracts, we rejected a total of 1,065 documents, selecting 336 for the second round. Then, we included 6 new papers found on Google Scholar. Finally, In the second round, we discarded a total of 25 documents.

Table 1 shows the results of each round, while the PRISMA diagram can be seen in Fig. 2 [10].

[10] The PRISMA Flow Diagram has been generated using Haddaway et al. (2022).

Table 1
Number of documents per stage of the systematic peer review. Note that in the last stage, a paper could be included in more than one category at the same time. The Total column takes into account the six papers retrieved from Google Scholar. Three of them talk about data and open data and the other three papers talk about new datasets published by their authors.

Step	Web of science	IEEE Xplore	Total
1. Query retrieve	995	405	1401
2. Screening	239	97	336
3. Included from Google Scholar	0	0	342
4. Manual tag	223	94	317
(a) Uses data	132	59	191
(b) Generates data	71	23	94
(c) Talks about data	33	7	40
(d) Talks about traceability	0	4	4
(e) Talks about lack of data	22	0	22

3. Analysis and discussion

3.1. Datasets collected

From the literature analysis detailed in the previous section, we have identified a total of 104 agriculture-related datasets from 196 papers. The diversity and heterogeneity of these data sources are significant, thus necessitating an initial analytical approach that involves categorizing the obtained results into distinct categories.

In the first place, it is important to note that unless otherwise stated, we are going to focus only on datasets that received, at least, two references from different papers. There are a lot of datasets with no citations, which correspond to datasets that are listed on the web page of the article because they are a result of it, but not mentioned in the paper. This situation is frequent in MDPI or IEEE papers) or with just one citation. The effort required to analyze every existing dataset will be out of the scope of a paper and would require creating a database or a website to catalog the information.

In the second place, Fig. 3 shows the categories of datasets we identified in our results, by the number of citations received. As we can see, most citations go to Satellite data sources, which received about 70% of the overall citations. These results show the very important impact that Satellite sources are having in the Agriculture field. Images datasets, which contain photos of plants, leaves, and insects are also a relevant source of information, grouping more than 12% of citations. Weather, Statistics, and GIS information were cited three times per category.

When it comes to publishers, there is a noticeable dominance of public organizations when it comes to publishing datasets with a higher number of citations. On the other hand, there is a comparatively small number of private organizations or individuals who choose to release their research data under open licenses or into the Public Domain.

The importance of Earth Observation data sources is explained by Donaldson and Storeygard (2016): Climate Change, topographic data, and agriculture and property policies have a relevant economic impact and require the use of data that, frequently, is provided by satellites. The authors also mention the M3 Crops Database, which combines public agriculture and satellite data. Turner et al. (2015) explains the importance of satellite data to preserve biodiversity conservation. as satellites are a very important tool for monitoring the status of biodiversity loss. It is also relevant to mention the work from Mukherjee et al. (2021) where the authors analyze the current prevalence of government satellite data, but draw a future where private data will become more predominant.

These results impose the need to go deep into the details of satellite sources, as is pretty clear that a large amount of papers use satellite data applied to the field of Agriculture. A Summary of satellite data

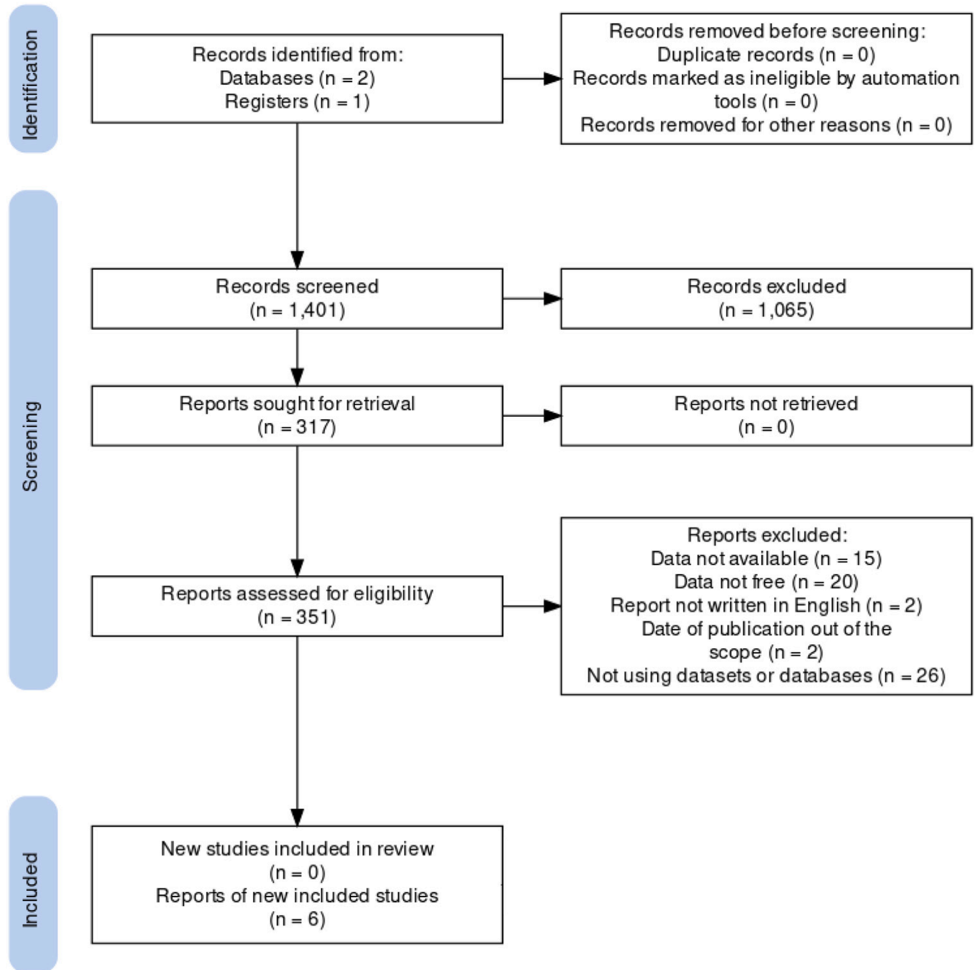

Fig. 2. PRISMA flow diagram.

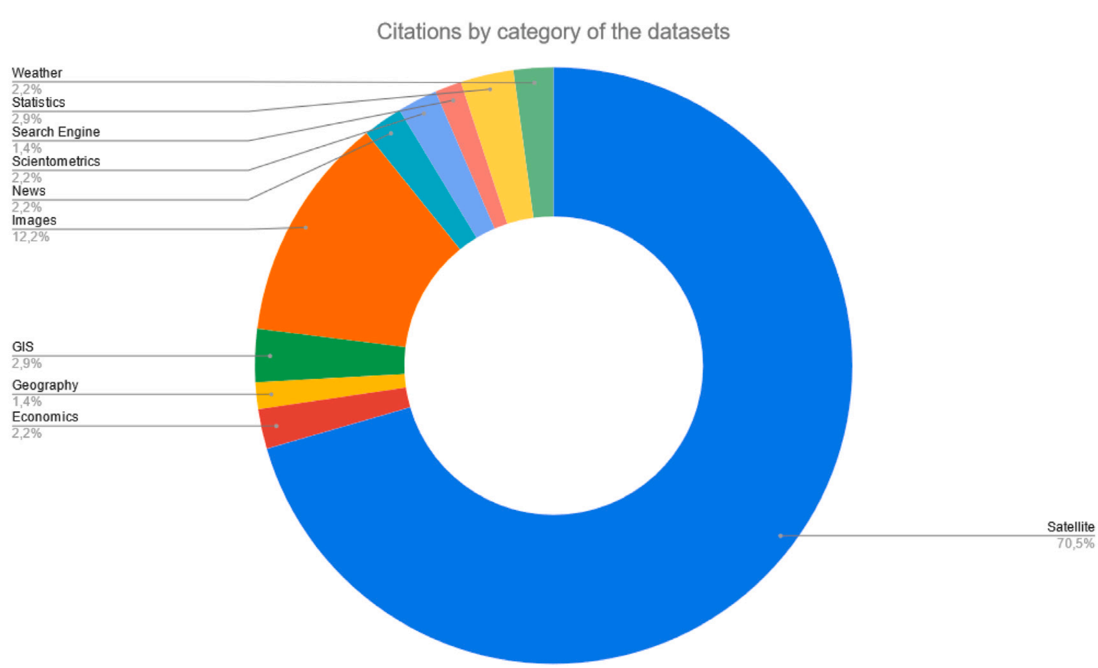

Fig. 3. Categories of datasets.

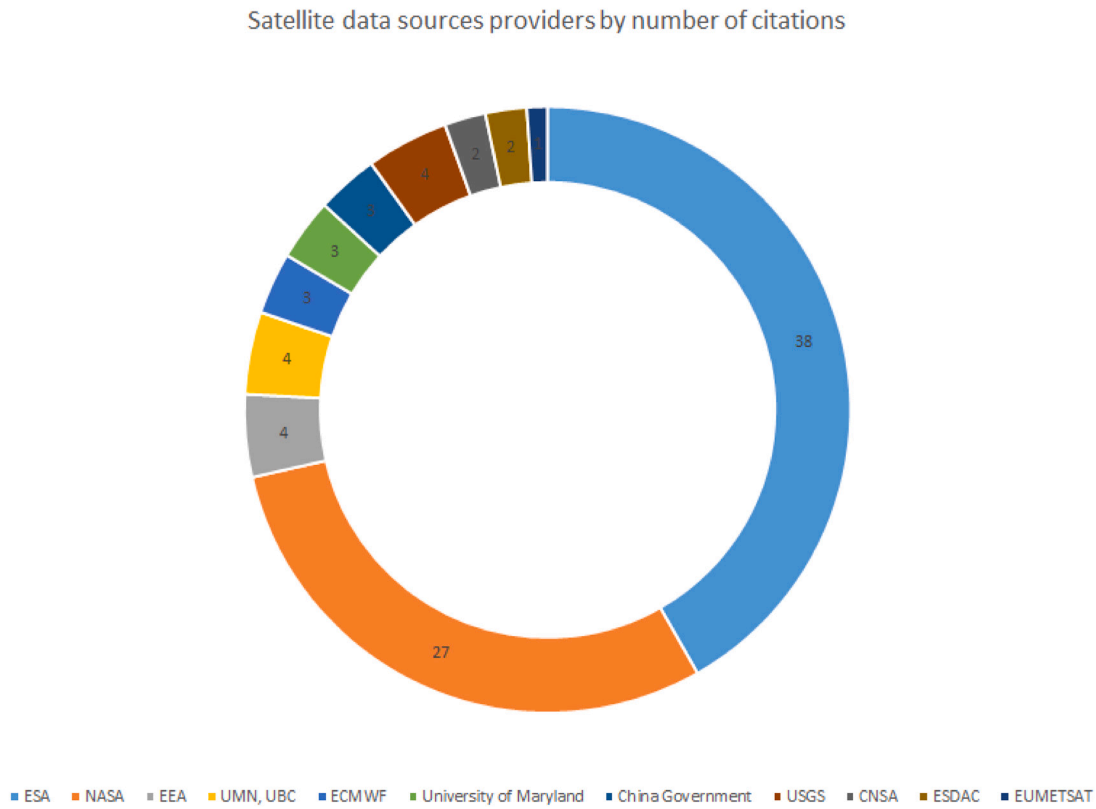

Fig. 4. Institutions (agencies, universities, governments, etc.) which provide datasets, by number of citations received. ESA: European Space Agency, NASA: National Aeronautics and Space Administration, EEA: European Environment Agency, ECMWF: European Centre for Medium-Range Weather Forecast, USGS: United States Geological Survey, CNSA: China National Space Administration, ESDAC: European Soil Database, EUMETSAT: European Organisation for the Exploitation of Meteorological Satellites, UMN: University of Minnesota, UBC: University of British Columbia.

sources can be shown in Table 2. Most data sources are provided by the European Space Agency (ESA) and by the National Aeronautics and Space Administration (NASA), which also concentrates most citations. Fig. 4 represents the number of citations received by each institution that published data sources. ESA concentrates a total of 38 citations, followed by NASA with 27 citations. Both agencies received more than half of the citations.

Sentinel is the most cited data source. Provided by ESA, Sentinel (European Space Agency, 2023) is a network of satellites with the goal of collecting data about agriculture, weather, climate, water, and soil. The first satellite was launched in 2014 and currently, the are six satellites in production. Sentinel mission is part of Copernicus; an Earth observation program of ESA (European Space Agency, 2018).

Concerning Moderate Resolution Imaging Spectrometers (MODIS) (NASA, 2023), this data source is composed of Terra and Aqua, which monitor the Earth's surface to improve our knowledge of processes that affect land, oceans, and atmosphere. MODIS is funded by NASA.

Apart from European and United States projects, China is also a data provider thanks to the Globeland30 dataset, which allows open access to data from 2000 and 2010. In addition, the Chinese satellite Gaofen-3, which was launched in August 2016, is also providing open access data (Sun et al., 2017).

Image datasets are also relevant data sources used in agriculture literature. In Table 3 we can see different data sources we have found in the collected data. The PlantVillage Dataset (Mohanty, 2016) is, by far, the most used data source in the papers we analyzed. This dataset contains a huge collection of healthy and unhealthy leaf images.

ImageNet (Yang et al., 2020b) is a visual database whose elements are organized according to WordNet Ontology. Unfortunately, ImageNet does not own the copyright of the images and we cannot consider Open Data.[11]

Furthermore, we would like to mention Table 4, which is a summary of statistical and economics data sources used in agriculture academic literature. FAOSTAT (Agriculture Organization of the United Nations, 2023) is the open data repository of the Food and Agriculture Organization of the United Nations and provides data on 245 countries from 1961. The World Bank (World Bank, 2023) is also a remarkable source used by authors who want to include statistical data in their research.

Finally, as shown by Fig. 3, there are more categories that we can mention, like Weather, Search engines, Scientometrics, or News. But we do not believe that a table for each one should provide relevant information. Anyhow, we can list some relevant facts. For example, AgInjuryNews[12] is a collection of Agriculture reports focused on work accidents. Is being used as a data source for papers that talks about accidents in the United States like Weichelt et al. (2022), Weichelt and Gorucu (2019), Gorucu et al. (2019). Web of Science and Google Search Scholar are also data sources cited by articles that perform literature reviews (Gu et al., 2020; Jones et al., 2021a; Chakraborty et al., 2017).

For countries and supra-national entities, the United States, the European Space Agency, and the European Union have created almost all the relevant data sources. The United States has created 12 datasets,

[11] According to ImageNet Terms and Conditions, https://www.image-net.org/about.php: *For researchers and educators who wish to use the images for non-commercial research and/or educational purposes, we can provide access through our site under certain conditions and terms*.

[12] https://aginjurynews.org/ (accessed on 16 April 2023).

Table 2
List of satellite data sources. Most satellite information comes from the European Space Agency or NASA and most of the data is publicly available under open data terms and conditions. The only exceptions found are ERA5, which requires registering and asking permission to download the data, and GlobCover, which allows using data for educational or research purposes but does not follow an open data policy. ESA: European Space Agency, NASA: National Aeronautics and Space Administration, EEA: European Environment Agency, ECMWF: European Centre for Medium-Range Weather Forecast, USGS: United States Geological Survey, CNSA: China National Space Administration, ESDAC: European Soil Database, EUMETSAT: European Organisation for the Exploitation of Meteorological Satellites, UMN: University of Minnesota, UBC: University of British Columbia. All links were accessed on 9 April 2023.

Dataset	Location	Citation	Publisher	Open Data	Citations	Cited by
Plant	https://sentinel.esa.int/web/sentinel/sentinel-data-access		ESA	Open Data	21	Zhang et al. (2022), Waleed et al. (2022), Singh and Das (2022), Tavus et al. (2022), Nuwarinda et al. (2021), Bukowiecki et al. (2021), Le et al. (2020), Zhao et al. (2016), Jiang et al. (2020), Sykas et al. (2022), Weikmann et al. (2021a), Del'Arco Sanches et al. (2018), Ranjbar et al. (2021), Christiansen et al. (2019), Khan et al. (2020), Liu et al. (2022a), Bengana and Heikkilä (2021), Xue et al. (2022), Tetteh et al. (2021), Chemura et al. (2017), Orti et al. (2021)
MODIS	https://modis.gsfc.nasa.gov/data/		NASA	Public Domain	17	Koley and Jeganathan (2023), Wassie et al. (2022), Waleed et al. (2022), Singh and Das (2022), Chen et al. (2021), Dutta et al. (2019), Dell'Acqua et al. (2018), Lu et al. (2017), Zhao et al. (2016), Nguyen et al. (2015), Vadrevu and Lasko (2015), De Clercq et al. (2015), Sheffield et al. (2015), Zhu et al. (2014), Biswas et al. (2015), Xue et al. (2022), Pérez-Hoyos et al. (2014)
Landsat	https://landsat.gsfc.nasa.gov/		NASA	Public Domain	5	Waleed et al. (2022), Amano and Iwasaki (2020), Souza-Filho et al. (2016), Singh et al. (2015), Jiang et al. (2020)
WorldView-2	https://earth.esa.int/eogateway/missions/worldview-2		ESA	Open Data	4	Chellasamy et al. (2014), Adjorlolo et al. (2014), Bengana and Heikkilä (2021), Chellasamy et al. (2015)
CORINE land cover	https://land.copernicus.eu/pan-european/corine-land-cover		EEA	Open Data	4	Kuleli and Bayazit (2022), Carles et al. (2017), Toth et al. (2013), Pérez-Hoyos et al. (2014)
EarthStat	http://www.earthstat.org/		UMN and UBC	Creative Commons Attribution 4.0	4	Liu et al. (2022b), Iizumi and Sakai (2020), Iizumi et al. (2018), d'Amour et al. (2017)
ERA5	https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5		ECMWF	Not open data	3	Bregaglio et al. (2022), Singh and Das (2022), Zhu et al. (2022), d'Amour et al. (2017)
Global Forest Change	https://glad.earthengine.app/view/global-forest-change		University of Maryland	Creative Commons Attribution 4.0	3	Kinnebrew et al. (2022), Pendrill and Persson (2017), Heino et al. (2015)
GlobCover	http://due.esrin.esa.int/page_globcover.php		ESA	Educational or research use only	3	Kinnebrew et al. (2022), Pendrill and Persson (2017), Heino et al. (2015)
Globeland30	http://globeland30.org/home_en.html		China Government	Open acces	3	Arsanjani (2018), Pendrill and Persson (2017), Lu et al. (2017)
National Land Cover Database (NLCD)	https://www.usgs.gov/centers/eros/science/national-land-cover-database		USGS	Public Domain	3	Cowger et al. (2019), Nuwarinda et al. (2021), Robertson and Saad (2013)
TerraSAR	https://earth.esa.int/eogateway/missions/terrasar-x-and-tandem-x		ESA	Creative Commons Attribution-ShareAlike 3.0 IGO	3	Zhao et al. (2014), Dumitru et al. (2015), Minh et al. (2019)
Gaofen	https://www.eoportal.org/satellite-missions/gaofen-3		CNSA	Open data	2	Jiang et al. (2020), Zhang and Chi (2020)

(*continued on next page*)

receiving 40.2% of the total citations, while the European Space Agency is responsible for 5 datasets (26.2% of citations) and the European Union has funded, created, or maintained 3 datasets receiving 8.2% of citations. Switzerland is another relevant player, with two datasets

Table 2 (*continued*).

Dataset	Location	Citation	Publisher	Open Data	Citations	Cited by
Land Cover	https://climate.esa.int/en/projects/land-cover/		ESA	Creative Commons Attribution-ShareAlike 3.0 IGO	2	Lu et al. (2017), Chemura et al. (2017)
LUCAS	https://esdac.jrc.ec.europa.eu/projects/lucas		ESDAC	Open data	2	Jimenez-Gonzalez et al. (2021), Toth et al. (2013)
Copernicus Open Access Hub	https://scihub.copernicus.eu		ESA	Open Access	1	Dell'Acqua et al. (2018)
MERRA-2	https://journals.ametsoc.org/collection/MERRA2		NASA	Open data	1	Lasko et al. (2018)
AgMERRA Climate Forcing Dataset	https://data.giss.nasa.gov/impacts/agmipcf/agmerra/		NASA	Public Domain	1	Ruane et al. (2014)
AIRSAR	https://airsar.jpl.nasa.gov/documents/dataformat.htm		NASA	Public Domain	1	Parikh et al. (2022)
ASCAT	https://www.eumetsat.int/ascat		EUMETSAT	Public Domain	1	Bykov et al. (2022)
COSMO-SkyMed	https://earth.esa.int/eogateway/missions/cosmo-skymed		ESA	Open data	1	Minh et al. (2019)
DISCover	https://data.giss.nasa.gov/landuse/		NASA	Public Domain	1	Yang et al. (2019)
DISTRIBUTED ACTIVE ARCHIVES	https://ladsweb.modaps.eosdis.nasa.gov/		NASA	Open data	1	Dell'Acqua et al. (2018)
EarthExplorer	https://earthexplorer.usgs.gov/		USGS	Use data from third parties	1	Pan et al. (2022)
EOS	https://eospso.gsfc.nasa.gov/		NASA	Public Domain	1	Dell'Acqua et al. (2018)
ESODIS	https://worldview.earthdata.nasa.gov/		NASA	Public Domain	1	Dell'Acqua et al. (2018)
Pleiades	https://earth.esa.int/eogateway/missions/pleiades		ESA	Open data	1	Bengana and Heikkilä (2021)
Radarsat	https://earth.esa.int/eogateway/missions/radarsat		ESA	Open data	1	Zhao et al. (2014)
SMAP	https://smap.jpl.nasa.gov/data/		NASA	Public Domain	1	Liu et al. (2020)
SMOS	https://earth.esa.int/eogateway/catalog/smos-science-products		ESA	Open data	1	Pacheco et al. (2015)

and 7.4% of the citations identified, the PlantVillage and the COCO datasets. More details can be found at Fig. 5.

An analysis of different strategies adopted by the United States and European Union around the open data can be found at Zuiderwijk and Janssen (2014). According to this work, in both institutions, open data policies have gained importance during the latest years. We can see this importance in terms of the different policies and agencies created. But at lower levels, there are still a lot of barriers that prevent the promotion of open data.

3.2. Open data and non-open data sources

As mentioned before in Section 1, according to the Budapest Initiative, an open-access data source must comply with the following characteristics (Budapest Open Access Initiative, 2023)[13]:

- Free availability on the public internet.
- Permitting any user to read, download, copy, distribute, print, search, or link to the data source.
- Use the data for any lawful purpose, without financial, legal, or technical barriers.

Being the only constraints the need to cite the original authors and to keep the integrity of their work. In our manuscript, we want to identify Open Data sources. Unfortunately, not every dataset collected from the literature can be considered as Open Data. Initially, we excluded datasets that require asking for permission to be downloaded.[14] But we found additional barriers. For example, some datasets do not include any mention of the usage license. Although these datasets are available for download on the Internet and the scientific community is already using them, if there is no mention of the usage conditions, we cannot consider them as Open Data sources. This is because, in some jurisdictions, if no explicit copyright or license terms are mentioned and the dataset is sufficiently original, the work is considered to have all rights reserved by default. While in other jurisdictions, especially the European Union, the reuse of the data might be restricted by sui generis database rights.[15]

Other barriers are the restriction of using the data for commercial purposes or only for scientific and educative reasons, which prevent the data from being considered Open Data.

[13] https://www.budapestopenaccessinitiative.org/faq/#openaccess (accessed on 16 April 2023).

[14] Though we decided to include in our analysis datasets which just require free online registration to be downloaded.

[15] Creative Commons and Data, https://wiki.creativecommons.org/wiki/Data (accessed on 25 February 2024).

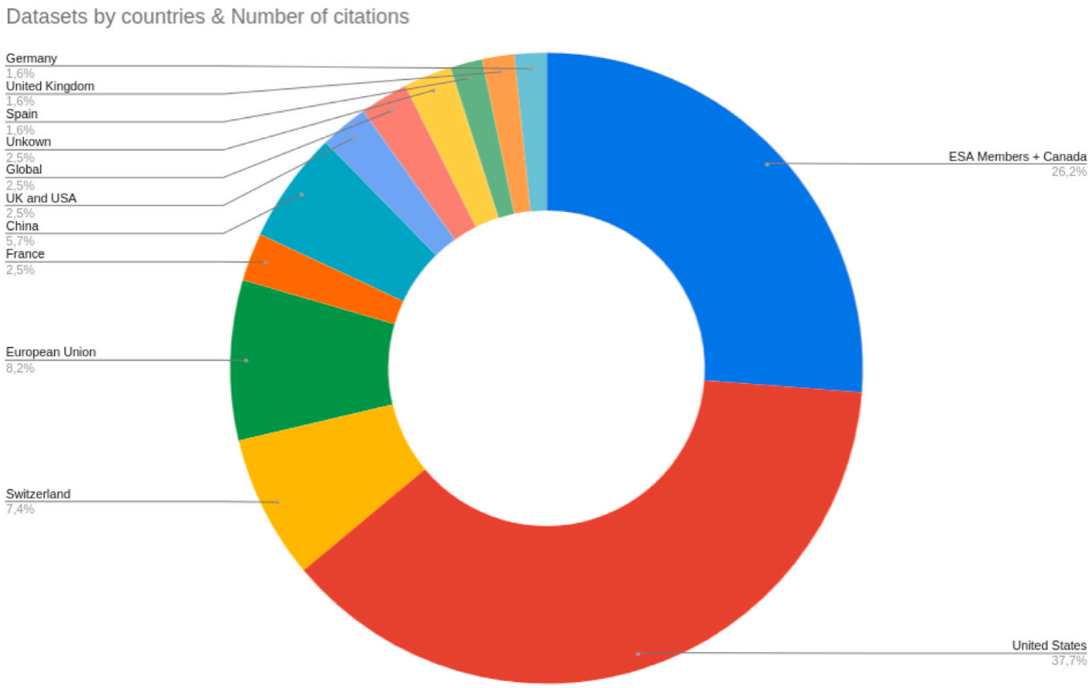

Fig. 5. Datasets per country, represented by the number of citations received. ESA Members: Austria, Belgium, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Luxembourg, the Netherlands, Norway, Poland, Portugal, Romania, Spain, Sweden, Switzerland, and the United Kingdom.

Table 3
List of images data sources. All links are accessed on 10 April 2023.

Dataset	Location	Citation	Publisher	Open Data	Citations	Cited by
PlantVillage Dataset	https://github.com/spMohanty/PlantVillage-Dataset		Sharada Mohanty	Creative Commons 1.0	8	Phan et al. (2022), Ahmad et al. (2023), Barburiceanu et al. (2021) Yang et al. (2020a) Khan et al. (2019) Vishnoi et al. (2023) Khattak et al. (2021) Ahmed et al. (2022)
ImageNet	https://www.image-net.org/		University of Princetown	Non-commercial use	5	Dac et al. (2022), Barburiceanu et al. (2021), Cowton et al. (2019) Yang et al. (2020a) Li et al. (2022c)
IP102	https://github.com/xpwu95/IP102	Wu et al. (2019)	Nankai University & Cardiff University	Unknown	3	Ren et al. (2019), Li et al. (2022c), Yang et al. (2021a)
COCO dataset	https://cocodataset.org/		COCO Consortium	Creative Commons 4.0	2	Dac et al. (2022), Dias et al. (2018b)
Corn disease and Severity	https://osf.io/s6ru5/	Ahmad et al. (2021)	Purdue University	Unknown	2	Phan et al. (2022), Ahmad et al. (2023)
Sugar Beets 2016	https://www.ipb.uni-bonn.de/data/sugarbeets2016/		University of Bonn	CC-BY-SA 4.0	2	Shorewala et al. (2021), Bosilj et al. (2018)

Finally, in addition to Open Data, we want to include Public Domain datasets despite being different concepts.[16] Public Domain sources are relevant in scientific research and have a positive impact that should be preserved or increased (Samuelson, 2003) so we wanted to consider this form of publishing creative work while not applying intellectual property, or using work whose rights have expired or are no longer applicable.

Fig. 6 illustrates all the characteristics that a data repository should follow to be included in our analysis of open data. We also consider Public Domain sources.

After checking the licenses of the most relevant datasets, we identified 11 data repositories that are not under the categories of Public Domain or Open data. The list of datasets is shown in Table 5. Even though all these datasets are available on the Internet for free, they have restrictions on the allowed use of the data. For example, if the datasets only allow the use of the data for academic, educational, or research purposes, then the dataset cannot be considered as Open Data. In other examples, like IP102 or the Corn Disease Severities dataset, the license is not provided. In some countries, if permissions are not explicitly defined by the author, it is considered that all permissions are reserved. In the case of ImageNet, the images are not the property of the authors from the dataset, so each image can have a different license. WorldView-2 deserves special attention: its license only provides Open

[16] https://openaccess.ox.ac.uk/2021/01/08/why-open-access-is-not-the-same-as-public-domain/ (accessed on 16 April 2023).

Table 4
List of economics and statistics data sources. All links are accessed on 14 April 2023.

Dataset	Location	Citation	Publisher	Open Data	Citations	Cited by
FAOSTAT	https://www.fao.org/faostat/		Food and Agriculture Organization of the United Nations	Free use data	5	Bykov et al. (2022), Hunink et al. (2012), Peltonen-Sainio et al. (2016) Yu et al. (2019), Liu et al. (2022b)
World Bank Open Data	https://data.worldbank.org		World Bank	Creative Commons Attribution 4.0 International	3	Zambrano-Monserrate et al. (2020), Montufar and Ayala (2019), Ayenew et al. (2018)
Statistical Data Warehouse	https://sdw.ecb.europa.eu/browse.do?node=9689727		European Central Bank	Open data & Non-Open data	1	Hoppe et al. (2014)

Table 5
List of repositories that are considered neither public domain or Open Data. P1: Free availability. P2: Permission to read, copy, distribute, search or link. P3: No restriction in the use of data.

Dataset	URL	#Cites	Public Domain	P1	P2	P3	Category	Comments
WorldView-2	https://earth.esa.int/eogateway/missions/worldview-2	5	No	No	No	No	Non-Open Data	https://earth.esa.int/eogateway/documents/20142/37627/WorldView-GeoEye-QuickBird-Terms-Of-Applicability.pdf/
Agin-juryNews	https://aginjurynews.org/	3	No	Yes	No	No	Non Open Data	https://aginjurynews.org/TermsOfUse/Index
ERA5	https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5	3	No	Yes	Yes	No	Non Open Data	https://ara.lmd.polytechnique.fr/index.php?page=tigr
GlobCover	http://due.esrin.esa.int/page_globcover.php	3	No	Yes	Yes	No	Non Open Data	http://due.esrin.esa.int/page_globcover.php
WorldClim	https://worldclim.org/	3	No	Yes	?	?	Non Open Data	Not explicitly
ARcGIS	https://www.arcgis.com/	2	No	Yes	No	No	Non Open Data	https://www.esri.com/en-us/legal/overview
Corn Disease and Severity	https://paperswithcode.com/dataset/cd-s	2	No	Yes	?	?	Non Open Data	Not explicitly
GLEAM	https://www.gleam.eu/	2	No	Yes	?	?	Non Open Data	Not explicitly
Google Scholar	https://scholar.google.com/	2	No	Yes	No	No	Non Open Data	~
ImageNet	https://www.image-net.org/	2	No	Yes	No	No	Non Open Data	https://www.image-net.org/about.php
IP102 dataset	https://github.com/xpwu95/IP102	2	No	Yes	?	?	Non Open Data	~

Data access for ESA member states, Canada and China. In this situation, we cannot consider WorlView-2 as Open Data.

A total of 29 citations are referred to non-open data repositories. This is 23% of the citations to the most relevant datasets.

3.3. Papers analysis

In the previous section, we analyzed the data repositories we found in the literature. At this point, it is relevant to extract some information regarding the papers using these data sources. Our goal is not to generate a list of papers but to provide some aggregated information regarding the papers which are citing the datasets.

To comprehensively cover the various subjects addressed by the authors in the plethora of papers we have encountered, our focus has been on analyzing the keywords provided by the authors themselves. Specifically, we have meticulously extracted papers that incorporate data derived from satellite sources, encompassing satellite imagery, economics, statistics, and related fields. This encompasses a wide range of data sources, including open data, public domain, and proprietary sources.

Table 6
Keywords by topics.

Category	Topic	Papers	Category	Topic	Papers	Category	Topic	Papers	Category	Topic	Papers
Satellite	Sensing	11	Satellite	SAR	2	Images	image	12	Economics & Statistics	soil	4
	Agriculture	11		Artificial	2		learning	10		nitrogen	2
	Remote	8		moisture	2		diseases	9		spatial	2
	resolution	7		Synthetic	2		feature	8		ecological	2
	mapping	7		measurements	2		extraction	7		moisture	4
	satellites	7		soil	2		deep	6		measurements	2
	segmentation	6		band	2		classification	6	Others	Agriculture	2
	learning	6		data	2		neural	6		injury	2
	Monitoring	4		networks	2		analysis	5			
	image	4		tobacco	2		convolutional	5			
	Spatial	4		stacking	2		color	4			
	aperture	4		accuracy	2		disease	4			
	radar	4		spatial	2		networks	4			
	Soil	4		land cover	2		support	4			
	Vegetation	4		random forest	2		vector	4			
	Benchmark	3		RF	2		machines	4			
	crop	3		erosion	2		agriculture	3			
	classification	3					segmentation	3			
	series	3					leaf	3			
	analysis	3					crops	3			
	Earth	3					architecture	3			
	remote	3					transfer	3			
	Image	3					detection	3			
	Sensors	2					modeling	3			
	Crops	2					diseases	3			
	fusion	2					identification	2			
	Reflectivity	2					plants	2			
	monitoring	2					training	2			
	Deep	2					computer	2			
	dataset	2					object	2			
	deep	2					texture	2			
	Time	2					lesions	2			
	Training	2					symptoms	2			
	Data	2					computational	2			
	Sentinel	2					network	2			
	synthetic	2					citrus	2			

To ensure the accuracy and relevance of our analysis, we have compiled a summary of the keywords associated with each topic, as depicted in Table 6. To filter out the noise and prioritize the most significant keywords, we have disregarded those with a frequency of occurrence below 2 so that minority topics are removed. Examining the table, it becomes evident that in the context of satellite sources, the most prevalent keywords revolve around Sensing and Agriculture. Furthermore, there is a diverse range of satellite-related keywords such as spatial, resolution, aperture, radar, earth, sentinel, and more. This fact provides an idea regarding the relevance of Satellite data in Agriculture research. Notably, field segmentation emerges as a prominent area of research among authors utilizing satellite datasets (Bengana and Heikkilä, 2021; Zhang et al., 2022; Sykas et al., 2022; Tetteh et al., 2021) as well as vegetation analysis (Ranjbar et al., 2021; Khan et al., 2020; Chemura et al., 2017), soil moisture (Singh and Das, 2022; Ranjbar et al., 2021), detecting damage caused by flood or pest (Zhang et al., 2022; Tavus et al., 2022) or machine learning tasks (Orti et al., 2021; Weikmann et al., 2021a; Sykas et al., 2022), among others. Fig. 7 shows a bar graph of authors' keywords from papers that use satellite datasets.

Concerning image data sources, agriculture is not one of the most used keywords, but image, learning and diseases occupy relevant positions. Papers which use image repositories tend to focus on deep learning (Phan et al., 2022; Ahmad et al., 2023; Barburiceanu et al., 2021; Vishnoi et al., 2023; Khattak et al., 2021; Ahmed et al., 2022), feature extraction (Yang et al., 2020a; Khan et al., 2019) and disease identification (Phan et al., 2022; Ahmad et al., 2023; Barburiceanu et al., 2021; Khan et al., 2019; Vishnoi et al., 2023). Economic and statistic sources are used mainly on soil analysis (Montufar and Ayala, 2019; Hoppe et al., 2014; Hunink et al., 2012; Bykov et al., 2022) or the effects of nitrogen contamination (Liu et al., 2022b; Hoppe et al., 2014). Other sources are used, for instance, to analyze labor injuries in the agriculture field (Weichelt et al., 2022; Weichelt and Gorucu, 2019; Gorucu et al., 2019).

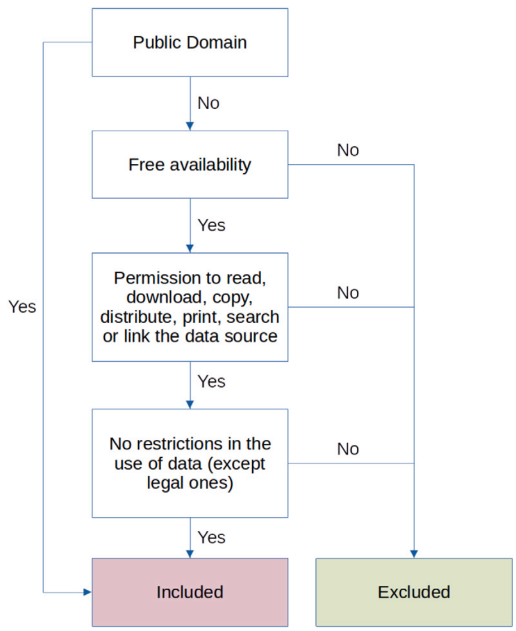

Fig. 6. Flow diagram to illustrate the steps that a dataset must follow to be included in our analysis as a Public Domain or Open Data repository.

As previously mentioned, we were taking into consideration all papers which cite data sources, without filtering those which are not Open Data. Taking into consideration Table 5, down below we list the number of papers that do not use public domain or open data repositories:

- Satellite: 15
- Image: 8
- Economy & statistics: 0
- Others: 11

Likewise, some papers use data repositories, there are published works from authors which generate data sources. From our papers' collection, we filtered papers that produce data from Web of Science and IEEE Xplore, we can identify a total of 100 papers that publish data at some level. The most frequent choice (50 papers) consists of publishing the data in the Supplementary Material section. Fig. 9 shows the distribution of different data repositories. A frequent choice by the authors is using a control version to publish their data. While the use of project o home webpages to advertise the data, keeping the files in webpages server is not a frequent choice. The use of specialized data repositories is also a frequent habit.

But, what about the license of the data? Concerning data published as supplementary data, it is hard to give an easy answer because it depends on the journal license, Most journals own the rights to their works and do not have Open Data or Public Domain license compatibility. so we can say that most of the supplementary material does not fit the license requirements. But we can focus on open-access options. For example, all MDPI journals publish their publication into a Creative Commons license which is compatible with Open Data.[17] Regarding IEEE, there are different licenses available. Unfortunately, most of them are not Open Data.[18] PLOS works are also open-data-compatible.[19] and the same happens with Nature Open Access papers[20] F1000Research articles are licensed under an open-data license too.

From our collection, we can consider that 37 papers publish their data as supplementary material that complies with Open Data requirements. For the rest of the repositories, we had to enter and download the data to check the license. Fig. 8 summarizes the results. Most repositories are Open Data, and Public domain repositories represent 10% of the total. There is a data source with different licenses, so we cannot consider it either open (and Public Domain) or non-open data. This said, it is important to mention the big differences found between IEEE Xplore and Web of Science collections, while most of the datasets found in articles retrieved by WoS are Open Data, the situation is completely different in the IEEE Collection where about 65% of datasets are Non-Open Data.

Table 7 show a list of Public Domain and Open Data repositories, linked to the work where the dataset was announced. For simplicity reasons, the table does not include datasets included in the Supplementary Data sections.

We found some articles whose authors use and generate data. It is not our intention to mention them one by one, but to illustrate this situation, we can mention, for example, Jones et al. (2021a), which is the paper related to the public domain repository (Jones et al., 2021b). The authors of this work use data from the Web of Science and Google Scholar.

3.4. FAIR principles

Publishing data is important to empower the research processes and science in general. However, the interoperability of this data is also a very important aspect that should be taken into consideration.

Different efforts are made to generate rich ecosystems of interoperable data. The FAIR Guiding Principles (Wilkinson et al., 2016) are a set of guidelines that aim to improve the use and management of scientific data. FAIR stands for "Findable, Accessible, Interoperable and Reusable".

- Findable makes references to the need that data would be easy to find. This implies the use of unique identifiers and that the data should be stored in liable repositories.
  - F1: (meta)data are assigned a globally unique and persistent identifier.
  - F2: data are described with rich metadata.
  - F3: metadata clearly and explicitly include the identifier of the data it describes.
  - F4: (meta)data are registered or indexed in a searchable resource.
- Accessible means that the scientific data must be available for free and without any barrier, except strictly needed ones.
  - A1: (meta)data are retrievable by their identifier using a standardized communications protocol.
  - A1.1: the protocol is open, free, and universally implementable
  - A1.2: the protocol allows for an authentication and authorization procedure, where necessary

[17] https://www.mdpi.com/authors/rights (accessed on 2 May 2023).

[18] https://www.ieee.org/publications/rights/open-access/index.html (accessed on 2 May 2023).

[19] https://journals.plos.org/plosone/s/licenses-and-copyright (accessed on 2 May 2023).

[20] https://www.nature.com/nature-portfolio/editorial-policies/self-archiving-and-license-to-publish#creative-commons-licences (accessed on 2 May 2023).

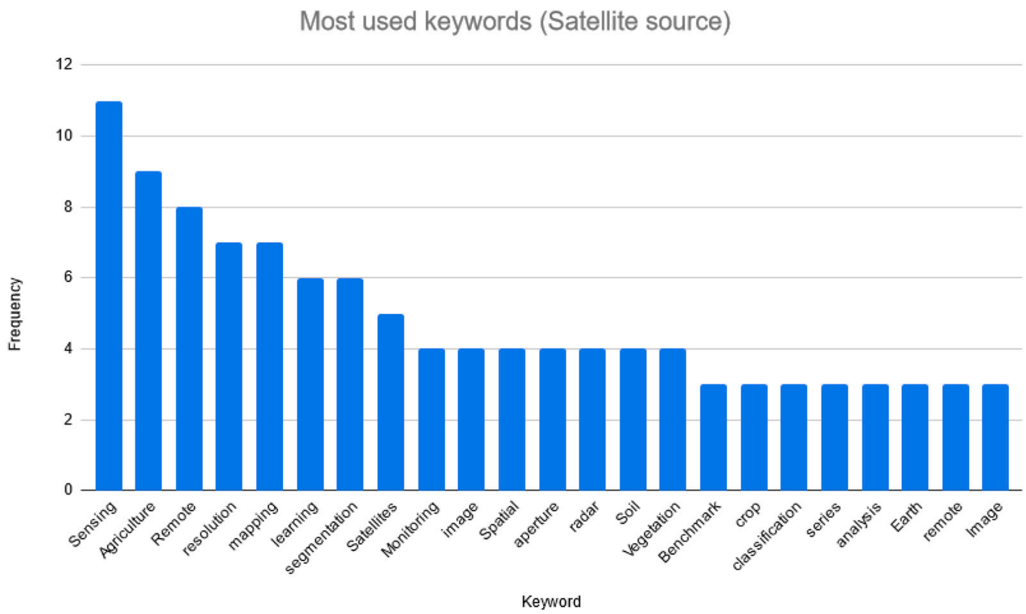

Fig. 7. Most relevant keyword used by papers that use satellite data sources. The bar graph only shows keywords with a frequency ≥2.

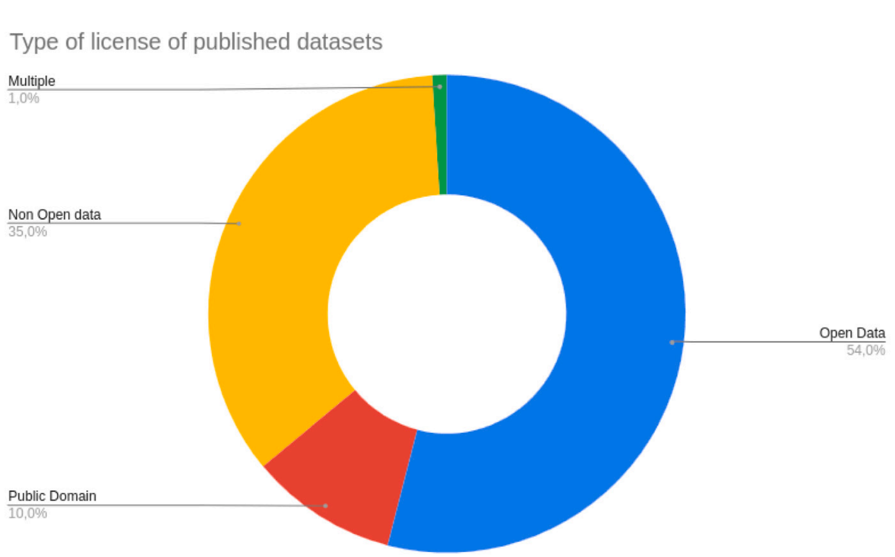

Fig. 8. Type of license of authors published datasets.

  – A2: metadata is accessible, even when the data are no longer available.

- Interoperable: applying metadata standards and common data formats is a requirement to make it possible to combine and relate the data.
  – I1: (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
  – I2: (meta)data use vocabularies that follow FAIR principles
  – I3: (meta)data include qualified references to other (meta) data
- Reusable: The data should be enough documented to make it easy to be reused by other scientists.
  – R1: (meta)data are richly described with a plurality of accurate and relevant attributes.
  – R1.1: (meta)data are released with a clear and accessible data usage license
  – R1.2: (meta)data are associated with detailed provenance
  – R1.3: (meta)data meet domain-relevant community standards

Together, the FAIR principles aim to improve the accessibility and utility of scientific data for the scientific community, which can in turn accelerate scientific progress and allow evidence-based decision-making.

To check the health status of the open data and public domain repositories, we want to analyze at which level the FAIR principles are being applied to the different repositories.

The FAIRmetrics Group aims to propose a develop metrics to check the level of compliance with the FAIR principles. As a result of the work done by this group, many tools, guidelines, and documentation have been published. A relevant example of their contribution is the

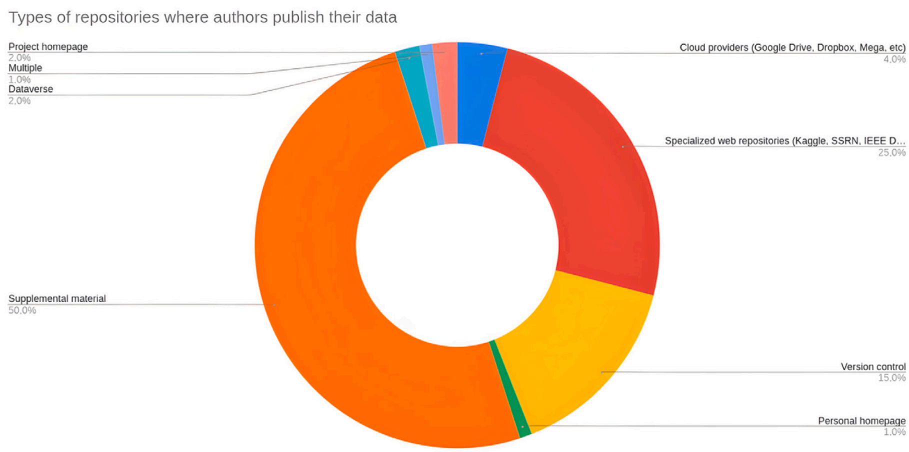

Fig. 9. This graph shows where the author publishes their data.

FAIRness Maturity Indicators and Tools.[21] Currently, there are two generations of maturity indicators. Gen1 indicators were proposed in Wilkinson et al. (2017b,a) and require some manual evaluation. Gen2 indicators can be checked automatically.

The assessment of Gen1 and Gen2 can be checked manually by survey-based evaluation (Wilkinson et al., 2017b) but also there are plenty of tools to assess automatically or partially automatically the compliance level with indicators. While Gen1 indicators should be checked manually, Gen2 can be analyzed by using automatic procedures.

One of these tools is the FAIR Evaluator,[22] launched by the FAIRmetrics Group. It is a service to evaluates the level of compliance of a resource. FAIR Evaluator supports MI tests, which are web APIs that check whether a resource complies with certain indicators. This API is being used by many services, for example, the FAIR Maturity Evaluation service,[23] which makes it easier to use the FAIR Evaluator API by providing an interactive user interface.

In this work, we want to analyze the degree of compliance with the FAIR principles of open data and public domain repositories generated by authors of Agriculture papers. Doing this task automatically allows us to process more repositories and prevent biases in the analysis. So we decided to use the API provided by the FAIR Evaluator service with the help of a Python script[24] implemented from scratch (see Table 8).

3.4.1. Compliance level

Table 9 describe the list of open data and public domain dataset and their degree of compliance with all the maturity levels studied in our work. In general, we can see that the datasets should improve mainly in Findability and Interoperability. None of the analyzed datasets complies with indicators Gen2_FM_FA2, RD-F4, Gen2_FM_I2B, Gen2_FM_I1 A, and RD-R1-3 while only one dataset complies with Gen2_FM_I1B. In general, the main problem is the lack of rich metadata which helps resources to be found and interpreted by machines.

3.4.2. Availability risks

Datasets are not only created, but also maintained and, sometimes, destroyed. In some circumstances, the deletion of data can be accidental and, in some other situations, intentional.

In any case, the availability of information is important to allow research processes to work. Thus, data availability is an important topic.

During the datasets' extraction phase, some datasets were unavailable. Some of them were temporarily unavailable, but most of them were permanently inaccessible.

Table 10 shows the list of datasets that are not available. The platform where the dataset is uploaded is very relevant, as most of these datasets were uploaded into project webpages. One of the datasets was pushed into a GitHub repository, that can be removed or moved by the authors. Three repositories were temporarily inaccessible in, at least, one of the three times we tried to access the data. In the case of Wagner et al. (2019), the data appeared as removed, but it was restored.

According to FAIR principle guidelines, maturity indicators Gen2_FM_F1 A and Gen2_FM_F1B are related to the availability of the resources. Concerning the first one, Gen2_FM_F1 A, all the analyzed datasets comply with the maturity indicator. Nevertheless, there were seven resources which do not comply with Gen2_FM_F1B. Except for (CSIC, 2019a), the rest of the resources are hosted on GitHub repositories. The main problem with GitHub is that repositories can disappear.

To prevent data loss, we have created a copy of each resource in Zenodo,[25] an open-source repository that is part of the OpenAIRE[26] project, funded by CERN and the European Commission. Zenodo provides every resource with a permanent identifier: a DOI. Meanwhile, is not possible to remove data uploaded into Zenodo, which is a reliable, well-known, and safe place to store open data. Zenodo also allows users to upload open-source datasets by preserving the author's rights. This is the main reason we decided to use Zenodo.

During the process of copying all the repositories into Zenodo, preserving the open license terms and giving proper credit to the author were priority objectives. All Zenodo's repositories were configured to preserve the same license and credit the authors. A brief note has been introduced to explain that the repository works as a backup of the original one. Table 11 shows the list of the five projects copied into Zenodo. Note that we have not migrated the repository from Li et al. (2022b) despite being Open Access (MIT License) because the authors are required to fill out a form with an institutional mail address before proceeding to download the data. We understand that the wish

[21] https://github.com/FAIRMetrics/Metrics/tree/master/MaturityIndicators (accessed on 3 May 2023).
[22] https://fairdata.services:7171/FAIR_Evaluator/ (accessed on 25 February 2024).
[23] https://fairsharing.github.io/ (accessed on 4 May 2023).
[24] The version of the script used in this work can be found published on Zenodo (Chamorro-Padial et al., 2024). Furthermore, later versions can be found in the project repository at https://github.com/rhizomik/fair-check.
[25] https://zenodo.org/ (accessed on 25 February 2024).
[26] https://www.openaire.eu/ (accessed on 25 February 2024).

Table 7
List of open data or public domain datasets created and published as data from the collected papers.

Dataset	Paper	Dataset cite	License
Campo Verde Database	Del'Arco Sanches et al. (2018)	Ieda Del'Arco Sanches (2017)	Open Data
Date Fruit Dataset for Automated Harvesting and Visual Yield Estimation	Altaheri et al. (2019b)	Altaheri et al. (2019a)	Open Data
Eyes in the Skies: A Data-driven Fusion Approach to Identifying Drug Crops from Remote Sensing Images Dataset	Ferreira et al. (2019)	Anselmo Ferreira (2018)	Open Data
Poultry barn mapping	Robinson et al. (2022a)	Caleb et al. (2023)	Open Data
Multi-species fruit flower detection using a refined semantic segmentation network	Dias et al. (2018b)	Dias et al. (2018a)	Public Domain
Real-time crop recognition in transplanted fields with prominent weed growth: a visual-attention-based approach	Li et al. (2019)	Nan et al. (2020c)	Open Data
TimeSen2Crop	Weikmann et al. (2021a)	Weikmann et al. (2021b)	Open Data
Varietal Classification of Rice Seeds Using RGB and Hyperspectral Images	Fabiyi et al. (2020)	Vu et al. (2019)	Open Data
A global database of diversified farming effects on biodiversity and yield	Jones et al. (2021a)	Jones et al. (2021b)	Public Domain
PLANT services plant book	Molina-Venegas et al. (2021)	Molina Venegas (2021)	Open Data
A global dataset for crop production under conventional tillage and no tillage systems	Su et al. (2021)	SU (2021)	Open Data
A global yield dataset for major lignocellulosic bioenergy crops based on field measurements	Li et al. (2018a)	Li et al. (2018b)	Public Domain
A new perspective when examining maize fertilizer nitrogen use efficiency, incrementally	Kitchen et al. (2022a)	Kitchen et al. (2022b)	Open Data
Resource-mediated indirect effects of grassland management on arthropod diversityResource-mediated indirect effects	Simon and Weisser (2017)	Simons et al. (2015)	Public Domain
Weed seedling images of species common to Manitoba, Canada	Beck et al. (2020a)	Beck et al. (2020b)	Public Domain
Digital CSIC	Delgado-Gonzalez et al. (2022)	CSIC (2019b)	Public Domain
MTOA	Li et al. (2022a)	Li et al. (2022b)	Open Data
DeepWeeds	Olsen et al. (2019a)	Olsen (2023)	Open Data
Platform for UAV Controlling	da Silva et al. (2022a)	Silva (2022)	Open Data
EasyPCC v2	Sommermann et al. (2018)	Guo et al. (2023)	Open Data

(*continued on next page*)

Table 7 (*continued*).

Long time series (2001–2015) high-resolution crop yield and water productivity dataset of China	Cheng et al. (2022)	Cheng (2021)	Open Data
Improved accuracy and reduced uncertainty in greenhouse gas inventories by refining the IPCC emission factor for direct N2O emissions from nitrogen inputs to managed soils	Hergoualc'h et al. (2021)	Hergoualc'h (2021)	Public Domain
Large-scale rollout of extension training in Bangladesh: Challenges and opportunities for gender-inclusive participation.	Medendorp et al. (2022)	Medendorp et al. (2022)	Open Data
Towards Fairer Datasets: Filtering and Balancing the Distribution of the People Subtree in the ImageNet	Williams et al. (2022)	Yasarer et al. (2021)	Public Domain
Severe recent decrease of adult body mass in a declining insectivorous bird population	Paquette et al. (2014)	Rioux Paquette et al. (2014)	Public Domain

of authors is to avoid a public download link. To respect their decision, we decided not to copy the data into a Zenodo repository.

We also skipped Delgado-Gonzalez et al. (2022), the authors use data from CSIC, a Spanish Government agency that requires registration before entering into their open data access portal.

3.5. Relevant topics in data management

During the literature analysis, we found not only papers that use or generate data but also papers that give interesting information about the lack of open data, barriers to the adoption of repositories, privacy concerns, or other topics related to data management.

In this section, we summarize different topics that, from our point of view, should be mentioned.

3.5.1. The use of research data

Some articles propose frameworks or strategies to improve the reusability or the performance of research data. A good example is Sangani et al. (2019), whose authors analyze the structure of datasets and their impact on the results of interpolation methods for mapping soil salinity. So it is important to find a suitable method according to the intrinsic characteristics of the dataset.

The rise of new technologies and the so-called "Agriculture 4.0" (Yang et al., 2021b) revealed the need for Smart Agriculture. We can mention, on one hand, the need for cybersecurity for datasets, as production data has weak security measures and is more vulnerable to data theft attacks than the data in the urban industry. On the other hand, the privacy of the information is also an important concern.

Data in heterogeneity should be handled to improve the feasibility of processing it from machines and to be able to combine different data sources. Ontologies can be defined as "abstract models of some aspect of the world, taking the form of a definition of the properties of important concepts and relationships" (Baader et al., 2004). Ontologies can contribute to enriching metadata and help machines to process complex knowledge. In this sense, we found some contributions that try to apply ontologies in Agriculture. Dumitru et al. (2015) use ontologies to model the information extracted from TerraSAR-X. Chukkapalli et al. (2020) applies ontologies to modelate farming collaboration. Mughal et al. (2021a) uses ontologies to define a conceptual model of river flow. More examples about applying ontologies can be found at Venkatesan et al. (2018), Harper et al. (2018).

The authors of Rathore et al. (2018) specifically talk about data management in the context of managing agriculture datasets.

3.5.2. Lack of research data

We can start by mentioning works where the lack of information is expressed by its authors. We found 22 papers where, at some point, it mentioned the need to have more research data. We do not want to mention all the articles, one by one. But we can summarize one of them. For example, Nedelciu et al. (2020) analyzes the problem of not having enough data in the field of Agriculture, about Phosphorus. The authors performed a literature review on different research databases and identified four areas where the lack of data is important. Moreover, the authors also analyze the consequences of not having open access to data.

Silva et al. (2022) makes a study about pesticides and their relation with human health. There is a subsection named *Pesticide use and risk* where the analysis is made using pesticide risk proxies, the authors explicitly mention the lack of monitoring data.

Booth and Kucharik (2022) analyze the equilibrium among manure nutrients and cropland demands. The authors explain the problem of the lack of data at the sub-county level (United States) and the accuracy problems caused by this absence of information.

The lack of data also affects innovation in the United States and in Europe, according to Cummings et al. (2021), Belletti et al. (2020). The first paper identified the lack of data as a barrier to the innovation of nano-agrifoods. The second one mentioned the lack of information as a "critical challenge" to study the river's biodiversity. The authors also stated the impact of having research data at different scales, which is closely connected with the problem stated by Booth and Kucharik (2022).

3.5.3. Blockchain and privacy

Blockchain can be a key technology to deal with privacy concerns and make it easier to share information giving fair remuneration to

Table 8
Description of FAIR Maturity Indicators used in our analysis. Source: https://fairsharing.github.io/FAIR-Evaluator-FrontEnd/#!/metrics (Accessed on 13 May 2023).

Indicator	FAIR Principle	Description
Gen2_FM_F1A	F1	Whether there is a scheme to uniquely identify the digital resource.
Gen2_FM_F1B	F1	The unique identifier of the data resource is likely to be persistent.
Gen2_FM_I3A	I3	Whether the linked data metadata contain links that are not from the same source (domain/host)
Gen2_FM_FA2	A2	Metric to test if the metadata contains a persistence policy, explicitly identified by a persistence Policy key (in hashed data) or a http://www.w3.org/2000/10/swap/pim/doc#persistencePolicy predicate in Linked Data.
Gen2_FM_R1.1	R1.1	Maturity Indicator to test if the linked data metadata contains an explicit pointer to the license. Tests: xhtml, dvia, dcterms, cc, data.gov.au, and Schema license predicates in linked data, and validates the value of those properties.
Gen2_FM_I2A	I2	Maturity Indicator to test if the linked data metadata uses terms that resolve. This tests only if they resolve, not if they resolve to FAIR data, therefore is a somewhat weak test.
Gen2_FM_I1B	I1	Maturity Indicator to test if the metadata uses a formal language broadly applicable for knowledge representation. This particular test takes a broad view of what defines a 'knowledge representation language'; in this evaluation, a knowledge representation language is interpreted as one in which terms are semantically grounded in ontologies. Any form of RDF will pass this test (including RDF that is automatically extracted by third-party parsers such as Apache Tika).
Gen2_FM_I2B	I2	Maturity Indicator to test if the linked data metadata uses terms that resolve to linked (FAIR) data.
Gen2_FM_I1A	I1	Maturity Indicator to test if the metadata uses a formal language broadly applicable for knowledge representation. This particular test takes a broad view of what defines a 'knowledge representation language'; in this evaluation, anything that can be represented as structured data will be accepted.
Gen2_FM_A1.2	A1.2	Test a discovered data GUID for the ability to implement authentication and authorization in its resolution protocol. Currently passes InChI Keys, DOIs, Handles, and URLs. It also searches the metadata for the Dublin Core 'accessRights' property, which may point to a document describing the data access process. Recognition of other identifiers will be added upon request by the community.
Gen2_FM_A1.1	A1.1	Data may be retrieved by an open and free protocol. Tests data GUID for its resolution protocol. Currently passes InChI Keys, DOIs, Handles, and URLs. Recognition of other identifiers will be added upon request by the community.
f4-searchable	F4	Search for existing metadata about the resource URI in data repositories, such as DataCite, RE3data.
RD-F4	F4	We extract the title property of the resource from the metadata document and check if the RD-specific search engine returns the metadata document of the resource that we are testing.
RD-R1–3	R1.3	A domain-specific test for metadata of resources in the Rare Disease domain. It tests if the metadata is structured conforming to the EJP RD DCAT-based metadata model. No failures in the ShEx validation of the metadata content of the resource against the EJP RD ShEx shapes, will be sufficient to pass the test.

those who generate the information, meanwhile, the traceability of data is ensured. Yang et al. (2021b) proposes the use of blockchain to improve the transparency of product information on the agricultural product supply chain.

Zhang et al. (2020) analyses the use of food safety traceability systems based on blockchain technologies.

Wu and Ma (2022) mentions how blockchain can help Chinese agriculture to improve in terms of sustainability, decentralization, independence, and security.

3.5.4. Towards the interoperability of Agriculture data

Finally, there are efforts to empower data sharing and improve the interoperability of information.

We can start by mentioning Cheviron et al. (2018), a platform funded by the French Research Infrastructure for Analysis and Experimentation on Ecosystems to host a database to empower open science on biochemistry.

AgBioData is a consortium of agricultural biological databases that embrace FAIR principles. In Harper et al. (2018), the authors wanted to provide guideline recommendations, and challenges to promote

Table 9
FAIR Maturity Indicators.

Dataset	reference	Gen2_FM_F1A	Gen2_FM_F1B	Gen2_FM_I3A	Gen2_FM_FA2	f4-searchable	Gen2_FM_R1.1	Gen2_FM_I2A	RD-F4	Gen2_FM_I1B	Gen2_FM_I2B	Gen2_FM_I1A	Gen2_FM_A1.2	Gen2_FM_A1.1	RD-R1-3
Campo Verde Database	Ieda Del'Arco Sanches (2017)	T	T	T	F	F	F	T	F	F	F	F	T	T	F
Date Fruit dataset...	Altaheri et al. (2019a)	T	T	T	F	F	T	T	F	F	F	F	T	T	F
Eyes in the skies...	Anselmo Ferreira (2018)	T	T	T	F	F	F	T	F	F	F	F	T	T	F
Poultry barn mapping	Caleb et al. (2023)	T	F	F	F	F	F	F	F	F	F	F	T	T	F
Multi-species fruit flower...	Dias et al. (2018a)	T	T	T	F	T	F	T	F	F	F	F	T	T	F
Real-time crop recognition in transplanted...	Nan et al. (2020c)	T	F	F	F	T	F	F	F	F	F	F	T	T	F
TimeSen2Crop	Weikmann et al. (2021b)	T	T	T	F	F	T	T	F	F	F	F	T	T	F
Varietal Classification of Rice Seeds...	Vu et al. (2019)	T	T	T	F	T	T	T	F	F	F	F	T	T	F
A global database of diversified...	Jones et al. (2021b)	T	T	T	F	T	F	T	F	F	F	F	T	T	F
PLANT services plant book	Molina Venegas (2021)	T	T	T	F	T	T	T	F	F	F	F	T	T	F
A global dataset for crop production under...	SU (2021)	T	F	F	F	F	F	F	F	F	F	F	T	T	F
A global yield dataset for major lignocellulosic...	Li et al. (2018b)	T	T	T	F	T	T	T	F	F	F	F	T	T	F
A new perspective when examining maize fertilizer...	Kitchen et al. (2022b)	T	T	T	F	F	F	F	F	F	F	F	T	T	F
Weed seedling images	Beck et al. (2020b)	T	T	T	F	T	T	T	F	F	F	F	T	T	F
Digital CSIC	CSIC (2019a)	T	T	T	F	T	F	F	F	F	F	F	T	T	F
MTOA	Li et al. (2022b)	T	F	F	F	T	F	F	F	F	F	F	T	T	F
DeepWeeds	Olsen (2023)	T	F	F	F	T	F	F	F	F	F	F	T	T	F
Platform for UAV Controlling	Silva (2022)	T	F	F	F	F	F	F	F	F	F	F	T	T	F
EasyPCC v2	Guo et al. (2023)	T	F	F	F	T	F	F	F	F	F	F	T	T	F
Long time series (2001-2015) high-resolution...	Cheng (2021)	T	T	T	F	T	T	T	F	F	F	F	T	T	F
Improved accuracy and reduced uncertainty in...	Hergoualc'h (2021)	T	T	T	F	F	F	T	F	F	T	F	T	T	F
Large-scale rollout of extension training in...	Medendorp et al. (2022)	T	T	T	F	T	F	T	F	F	F	F	T	T	F
Towards Fairer Datasets: Filtering and...	Yasarer et al. (2021)	T	T	T	F	T	F	T	F	F	F	F	T	T	F
Severe recent decrease of adult body mass...	Rioux Paquette et al. (2014)	T	T	T	F	F	T	T	F	F	F	F	T	T	F

Table 10
List of datasets temporary or permanently unavailable. All datasets were accessed three times (on 13 March, 12 April and 14 May 2023).

Dataset name or URI	Paper	Comment	Published on
Queensland Suicide Register	Arnautovska et al. (2015)	Temporary non accesible	Project webpage
National Coroners Information System	Arnautovska et al. (2015)	Temporary non accesible	Project webpage
traps.ncipmc.org	Rosenheim and Gratton (2017)		Project webpage
Soybean dataset	Rosenheim and Gratton (2017)		Project webpage
European Pollen database	Fyfe et al. (2015)		Project webpage
https://www.naturalearthdta.com/	Bleasdale et al. (2020)		Project webpage
https://portal.mtt.fi/portal/page/portal/kasper/pelto/peltopalvelut/lajikekokeet	Peltonen-Sainio et al. (2016)		Personal homepage
https://nadp.slh.wisc.edu/MDN/maps.aspx	Yu et al. (2019)		Project webpage
https://www.pbl.nl/en/publications/2006/N2OAndNOEmissionFrom	Philibert et al. (2012)		Project webpage
https://github.com/liuliu66/AgriPest	Wu and Ma (2022)		GitHub
Sen4Agrinet	Sykas et al. (2022)		Project webpage
US county-level agricultural crop production typology	Wagner et al. (2019)	Temporary removed	FigShare

Table 11
List of datasets copied into Zenodo.

Repository	Original reference	Zenodo reference
Poultry barn mapping	Caleb et al. (2023)	Robinson et al. (2022b). Robinson et al. (2022c)
Real-time crop recognition in transplanted fields with prominent weed growth: avisual-attention-based approach	Nan et al. (2020c)	Nan et al. (2020a), Nan et al. (2020b)
DeepWeeds	Olsen (2023)	Olsen et al. (2019b)
Platform for UAV Controlling	Silva (2022)	da Silva et al. (2022b)
EasyPCC v2	Guo et al. (2023)	Guo et al. (2020)

good practices for sharing interoperable data. This includes ontology handling, metadata, and data curation.

Venkatesan et al. (2018) presents a knowledge-based system relying on the semantic web to integrate data about plant species. The system is named Agronomic Linked Data (AgroLD) and integrates 50 datasets.

López-Morales et al. (2020) Discuss the need to create interoperable systems, have standard interfaces, and make it easier to integrate different solutions. The authors study the creation of a platform for irrigation community management. Another example of a platform can be found at Wang et al. (2020)

Finally, we can mention Environmental Cloud for the Benefit of Agriculture (CEBA) (Sarramia et al., 2022), a data lake to store heterogeneous scientific data related to the environment.

4. Conclusions

In this study, we have downloaded and compiled information from numerous scientific articles related to Agriculture, following the PRISMA methodology. We focused on articles that utilize datasets or generate datasets.

Without delving into the analysis of open or public domain datasets yet, we found a strong prevalence of datasets containing satellite information. Seven out of ten dataset citations are directed towards this category of data repository. Datasets containing images follow closely behind, with a significant percentage (around 13%).

Satellite datasets are primarily published by public and governmental agencies, which explains why they also receive the majority of citations. Although we found more references to datasets from NASA than ESA, the latter receives a higher number of citations, as observed in Figs. 4 and 5.

The current analysis of academic literature makes it clear that we face significant challenges today. On the one hand, there is an almost global monopoly of information derived from satellite data compared to other types of information. While satellite data tends to be freely accessible by default, this situation is not observed in data from other sources. Various consortia and organizations are producing open agricultural data, but there is a need for greater awareness on the part of both the public and private sectors regarding the importance of circulating data openly. Is required to establish long-term projects and interdisciplinary approaches to improve the open data adoption and the use of Open Data and provide stakeholders with good reasons to adopt these technologies. As mentioned by Osinga et al. (2022), applied to the field of Big Data. In addition, more attention should be put to developing benchmarks to evaluate Open Data adoption (Susha et al., 2015). Finally, we recommend *The State of Open Data* for years 2022 and 2023 (Science et al., 2022, 2023), a report that presents the current status of Open Data in the world and discusses different strategies to facilitate the adoption of Open Data.

On the other hand, there is a need to promote access and data production globally. These efforts must be carried out both by public initiatives, promoting platforms and enabling legal and regulatory changes that support the dissemination of open data, and by private initiatives, collaborating on data exchange and defining standards that promote interoperability. Special attention should be placed on Latin America, Asia, and, particularly, Africa (Mehrabi et al., 2020; Nobes and Harris, 2023). Despite initiatives in these continents, they are insufficient or fail to reach small-scale farmers who sometimes struggle to access the internet, especially in the African context. In these cases, additional investments are necessary to provide these small-scale producers with the necessary technology to become consumers of this data.

The issue of internet access for small farmers falls outside the scope of this article. However, we cannot overlook the fact that it is a barrier that hinders the use and production of data related to agriculture. As demonstrated by the analysis of Mehrabi et al. (2020), between 24 and 37% of small farms (less than one hectare) have internet access through modern technologies (at least, access to a 3G mobile network). Meanwhile, in Africa, difficulties in accessing the internet are very important. The need to invest in improving internet access for these communities becomes evident.

While there are multiple initiatives and platforms, it is essential to stop working in isolation and unite efforts to continue advancing the development of open data platforms that facilitate interoperability.

We also need to highlight the role of the scientific community in this task. As we have seen in this article in Section 3.4, there are collections of data resulting from scientific efforts that are not adequately protected and could disappear. Therefore, we must make a special effort to preserve this information by publishing it on platforms known to the academic community that enable its preservation. We also need to progressively move towards improving the interoperability and re-usability of this data by applying the FAIR principles to them. Additionally, most of the recommendations from the report *The Budapest Open Access Initiative: 20th Anniversary Recommendations* (Budapest Open Access Initiative, 2022) refer to practices that involve the whole research community: researchers, journals and publishers.

Moving on to the main objective of this study, which is the analysis of open data sources, over half of the analyzed datasets have an open license, with an additional 10% of datasets published under the public domain. However, we encountered up to 35% of papers that are not openly available. It is worth mentioning that datasets that do not explicitly mention an open license are considered non-open, both as a precaution and due to legal requirements in different countries.

Regarding open datasets, we are particularly interested in assessing their compliance with the FAIR principles. To accomplish this, we developed Fair-check, a Python script that allows us to automatically check the degree of compliance with the FAIR principles' maturity indicators for a set of resources. At this point, we found that significant efforts are still needed to achieve optimal maturity, especially concerning interoperability, as shown in Table 9.

Another contribution made in this article is to increase the availability (Findability principle) of open datasets. For this purpose, we have republished all datasets that had not fulfilled the Findability indicators on Zenodo. As a result, all analyzed open data repositories now comply with these indicators.

Finally, we have dedicated space to other articles that, although not utilizing or generating datasets, discuss them. We found various topics of debate, which can be summarized as follows: utilization of research data, lack of research data, blockchain and privacy, and strategies to achieve interoperability.

5. Materials

All the references used, organized according to the different phases shown in Table 1, have been published on Zenodo to facilitate the reuse of our research data by the scientific community.

Furthermore, the software generated for the analysis of the FAIR principles in Section 3.4, which we named Fair-Check, can be obtained in two ways. Firstly, we have created a repository on GitHub from which the code is accessible, and we invite anyone interested to contribute. The repository is located at https://github.com/jorgechp/fair-check under a GNU GPL v3 license. Secondly, to preserve the generated code against both intentional or accidental deletion, a version has been published on Zenodo as of May 31, 2023.

Finally, the collection of papers used for this work is publicly available at Zenodo (Chamorro-Padial et al., 2023).

CRediT authorship contribution statement

Jorge Chamorro-Padial: Conceptualization, Data curation, Investigation, Resources, Software, Visualization, Writing – original draft, Writing – review & editing. **Roberto García:** Conceptualization, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing, Funding acquisition. **Rosa Gil:** Conceptualization, Funding acquisition, Project administration, Validation, Visualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jorge Chamorro-Padial reports administrative support and article publishing charges were provided by University of Lleida. Jorge Chamorro-Padial reports a relationship with University of Lleida that includes: employment and funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data is publicly available at Zenodo and is referenced in the article.

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